

Lippia sidoides and *Myracrodruon urundeuva* gel prevents alveolar bone resorption in experimental periodontitis in rats

M.A. Botelho^{a,b,*}, V.S. Rao^b, C.B.M. Carvalho^c, J.G. Bezerra-Filho^a, S.G.C. Fonseca^d,
M.L. Vale^b, D. Montenegro^g, F. Cunha^e, R.A. Ribeiro^b, G.A. Brito^{f,b}

^a Department of Community Health, Faculty of Medicine, Federal University of Ceará, Fortaleza, Brazil

^b Department of Physiology and Pharmacology, Federal University of Ceará, Fortaleza, Brazil

^c Department of Pathology, Faculty of Medicine, Federal University of Ceará, Fortaleza, Brazil

^d Department of Pharmacy, Federal University of Ceará, Fortaleza, Brazil

^e Department of Pharmacology, Federal University of Rio de Janeiro, São Paulo, Brazil

^f Department of Morphology, Federal University of Ceará, Fortaleza, Brazil

^g Christus Biological Research Center, Faculty of Medicine, Fortaleza, Brazil

Received 26 February 2007; received in revised form 24 June 2007; accepted 1 July 2007

Available online 14 July 2007

Abstract

In Brazilian folk medicine, *Lippia sidoides* (*Ls*) and *Myracrodruon urundeuva* (*Mu*) have gained popularity and reputation as effective antimicrobial and anti-inflammatory agents. This work aimed to evaluate the effect of topical herbal gel from *Ls* 0.5% (v/w) and *Mu* 5% (w/w) in experimental periodontal disease (EPD) in rats. Wistar rats were subjected to ligature placement around the second upper left molars. Animals were treated topically with *Ls* and/or *Mu*-based gel, immediately after EPD induction and three times/day for 11 days until the rats were sacrificed (11th day). Saline-based gel was utilized as control for all experiments and doxycycline based gel 10% (w/w) was utilized as reference substance. Animals were weighed daily. Alveolar bone loss was measured as the difference (in millimeters) between the cusp tip and the alveolar bone. The periodontum and the surrounding gingivae were examined at histopathology, as well as the neutrophil influx into the gingivae was assayed using myeloperoxidase activity and cytokine production mainly tumor necrosis factor- α (TNF- α) and interleukin-1 β (IL-1 β) levels by ELISA method. The local bacterial flora was assessed through culture of the gingival tissue in standard aerobic and anaerobic media. Alveolar bone loss was significantly inhibited by *Ls* and *Mu* combined treatment compared to the saline control group. *Ls* and *Mu* combined treatment reduced tissue lesion at histopathology, with partial preservation of the periodontum, coupled to decreased myeloperoxidase activity as well as significantly inhibited TNF- α and IL-1 β production in gingival tissue compared to the saline control group. *Ls* and *Mu* combined treatment also prevented the growth of oral microorganisms and the weight loss. *Ls* and *Mu* combined based gel treatment preserved alveolar bone resorption and demonstrated anti-inflammatory and antibacterial activities in experimental periodontitis.

© 2007 Elsevier Ireland Ltd. All rights reserved.

Keywords: Essential oil; *Lippia sidoides*; *Myracrodruon urundeuva*; Periodontitis; Bone resorption; Antimicrobial activity

1. Introduction

Periodontitis, a relevant cause of tooth loss in adults (Chambrone and Chambrone, 2006), is a chronic inflammatory disease that is characterized by localized bone resorption (Sallay

et al., 1982; Samejima et al., 1990). The pathogenesis of periodontitis involves the presence of a bacterial plaque that may initiate a local inflammatory reaction in predisposed hosts. The host inflammatory response leads to edema, leukocyte infiltration and release of inflammatory mediators, causing periodontal pocket formation, connective tissue detachment and, alveolar bone resorption, ultimately leading to tooth loss (Samejima et al., 1990; Menezes et al., 2005).

Lippia sidoides Cham. (Verbenaceae), popularly known as “alecrim pimenta” is largely grown in Northeast Brazil. The leaves yield an essential oil (EO) rich in thymol, which

* Corresponding author at: Departamento de Saúde Comunitária, Faculdade de Medicina, Universidade Federal do Ceará, Rua Professor Costa Mendes, 1608-5^o Andar, 60.430-270 Fortaleza, Ceará, Brazil. Tel.: +55 85 3366 8044; fax: +55 85 3366 8050.

E-mail address: marcobotelho1@gmail.com (M.A. Botelho).

is antimicrobial (Lemos et al., 1990). The hydro-alcoholic extracts of this plant are largely utilized for treatment of skin wounds, as a mouth antiseptic and in liquid soap preparations to treat and prevent general fungal infections of the body (Lacoste et al., 1996). A mouth-rinse prepared with the EO decreased *Streptococcus mutans* counts (Botelho and Soares, 1994) and in a double blind study, effectively reduced gingivitis in humans (Botelho, 2005), confirming the ethnic and traditional use of *Lippia sidoides* leaves to cure dental disorders (Matos, 2000).

The antibacterial activity of *alecrim pimenta* is known for decades and recently, its EO has been evaluated against *Candida albicans* and cariogenic bacteria (Botelho et al., 2007). Further, EO has been shown to exert topical anti-inflammatory, antioxidant, gastroprotective effects and low toxicity (M.V. Monteiro et al., 2006; Fontenelle et al., 2007).

Myracrodruon urundeuva (Engl.), popularly known as “aroeira do sertão”, is a medicinal plant traditionally utilized for treating bleeding gums and gynecological disorders (Viana et al., 2003; J.M. Monteiro et al., 2006). It is a large tree that also grows in Northeastern region of Brazil. Previous studies have shown that the hydro alcoholic extracts from *Myracrodruon urundeuva* stem bark exert anti-inflammatory activity in several experimental models of inflammation, such as carrageenan-induced paw edema, neutrophil migration to peritoneal cavity and hemorrhagic cystitis induced by cyclophosphamide (Viana et al., 1997, 2003). Analgesic, antiulcer, hepatoprotective, antidiarrheal and colonic anastomotic wound healing properties have been also reported (Viana et al., 1997; Goes et al., 2005). A chemical study has demonstrated that the ethyl acetate extract has two main fractions. One of them has predominance of substances of a chalcone nature and the other has predominance of catechic tannins (Viana et al., 1997).

Based on this background evidence, the present study was aimed to evaluate the effects of *Lippia sidoides* (*Ls*) and *Myracrodruon urundeuva* (*Mu*) based gels, separately or in combination, for a possible synergistic effect, in the rat model of experimental periodontal disease (EPD). This evaluation was based on the histopathological analysis, neutrophil influx, pro-inflammatory cytokine production and microbiological analysis.

2. Materials and methods

2.1. Plant material

Initially, Federal University of Ceará Ethical Committee (#551/04; #228/06) authorized the collection of the plants as follows: the leaves of *Lippia sidoides* were collected in August 2004 at the Medicinal Plants Garden by Prof. Francisco José Abreu Matos and the *Myracrodruon urundeuva* ground bark was collected at the outskirts of the city of Fortaleza, state of Ceará, Brazil. Taxonomic identification of the plants was performed by botanists of the Prisco Bezerra Herbarium, Department of Biology, Federal University of Ceará, where voucher specimen are deposited under the No. 25149 and 14999, respectively. *Lippia sidoides* leaf essential oil was extracted by using a modified Clevenger apparatus by the steam distillation technique (Craveiro et al., 1976). The essential oil was extracted from fresh leaves

(300 g) for 2 h. The oil was dried over anhydrous sodium sulphate, yielding 1.2% (w/w).

2.2. Gel preparation

Myracrodruon urundeuva 5% (w/w) gel and *Lippia sidoides* essential oil 0.5% (v/w) gel were prepared at the Faculty of Pharmacy, Federal University of Ceará. The fluid extract from ground bark of *Myracrodruon urundeuva* Fr. All. was prepared by maceration with glycerin–hydro-alcoholic solution and then percolated with hydro-alcoholic solution, then it was filtered through a muslin cloth for coarse residue and finally through Whatman No. 1 filter paper, measured and kept in an air tight amber colored container, the flask was kept under dark to avoid effect of light on the active ingredients of the *Myracrodruon urundeuva*. The carbopol-940 (BF Goodrich Co., Cleveland) 2% (w/v) was utilized by mechanical dispersion in distilled water under vigorous agitation for the preparation of both herbal gels. For the preparation of the *Myracrodruon urundeuva* gel at 5% (w/w), 50 ml of carbopol-940 and 2.5 ml of fluid extract of *Myracrodruon urundeuva* was utilized, being neutralized until pH 6.0 with trietanolamine. The *Lippia sidoides* 0.5% gel was prepared with 50 ml of carbopol-940, 0.25 ml of *Lippia sidoides* essential oil and 5 ml polysorbate 80, being neutralized until pH 6.0 with trietanolamine. The combined gel of *Myracrodruon urundeuva* extract and *Lippia sidoides* essential oil was prepared and the mixture contains 5% of *Mu* and 0.5% of *Ls*, respectively. The gels were stored in white polyethylene containers kept hermetically sealed under refrigeration at 8 °C until utilization.

Stability study to evaluate the consistency of the gel over a period of 2 months was conducted by keeping the formulation at different conditions (4 °C; 37 °C and room temperature 25 °C) and measuring the viscosity of the gel formulation at regular intervals. The viscosity was measured by Brooke-field synchroelectric viscometer. The TD bar spindle of LV series was employed for the measurement. Other parameters like sinresis (separation), color, moisture content, clarity and crystallization were also evaluated. The study indicated that the evaluated parameters of the carbopol gel did not change significantly throughout the period in the specified conditions.

2.3. Animals

One hundred and forty-four male Wistar rats (160–200 g) from our own animal facilities were housed in temperature-controlled rooms and received water and food ad libitum. All experiments were conducted in accordance with the local guidelines on the welfare of experimental animals and with the approval of the Committee of Ethics in Animal Research of the Federal University of Ceará.

2.4. Induction of experimental periodontal disease (EPD)

A sterilized nylon (000) thread ligature was placed around the cervix of the second left upper molar of rats anesthetized with 10% chloral hydrate (400 mg/kg, i.p.), as described elsewhere

(Menezes et al., 2005). The ligature was knotted on the buccal side of the tooth, resulting in subgingival position palatally and in supragingival position buccally. The contralateral right side was utilized as the unligated control (Menezes et al., 2005). Animals were weighed daily.

2.5. Drug treatments

The experiments were performed utilizing six animals per group. The experimental groups were divided in Naïve group (animals non-treated and not subjected to EPD), non-treated group (animals subjected to EPD which received no treatment), vehicle-treated group (animals subjected to EPD and treated with topical vehicle gel), *Mu* group (animals subjected to EPD and treated with topical *Myracrodruon urundeuva* 5%, w/w, gel), *Ls* group (animals subjected to EPD and treated with topical *Lippia sidoides* 0.5%, v/w, gel), *Mu + Ls* group (animals subjected to EPD and treated with topical *Myracrodruon urundeuva* 5% and *Lippia sidoides* 0.5% combined based gel) and DOX group (animals subjected to EPD and treated with topical 10%, w/w, doxycycline based gel (Pfizer, Karlsruhe, Germany)) which was utilized as a reference drug group. The topical treatment with the gel was performed immediately after the surgical procedure and three times/day until the sacrifice on the 11th day.

2.6. Measurements of alveolar bone loss

The animals were sacrificed on the 11th day of periodontitis induction, and had their maxillae excised and fixed in 10% neutral formalin. Both maxillary halves were then defleshed and stained with aqueous methylene blue (1%) in order to differentiate bone from teeth. The horizontal alveolar bone loss, the distance between the cusp tip and the alveolar bone, was measured using a modification of the method of Crawford et al. (1978) as described by Samejima et al. (1990). Measurements were made along the axis of each root of the first (three roots), second and third molar teeth (two roots). The total alveolar bone loss was obtained by taking the sum of the recordings from buccal tooth surface and subtracting the value of the right maxilla (unligated control) from the left, in millimeters (Leitão et al., 2005).

2.7. Histopathological analysis

After sacrifice, under anesthesia, animals had their maxillae excised. The specimens were fixed in 10% neutral buffered formalin and demineralized in 7% nitric acid. These specimens were then dehydrated, embedded in paraffin, and sectioned along the molars in a mesio-distal plane, for hematoxylin and eosin staining. Sections of 6 μm thickness, which included the roots of the first and second molars, were utilized. The areas between the first and second molars, where the ligature was placed, were analyzed under light microscopy using on a “0” (zero) to 3 score grade, considering the inflammatory cell influx, and alveolar bone and cementum integrity, as described previously (Menezes et al., 2005). Score “0” (zero): absence or only a discrete cellular infiltration (inflammatory cell infiltration is sparse and restricted to the region of the marginal gingival), preserved

alveolar process and cementum. Score 1: moderate cellular infiltration (inflammatory cellular infiltration present all over the insert gingival), some but minor alveolar process resorption and intact cementum. Score 2: accentuated cellular infiltration (inflammatory cellular infiltration present in both gingival and periodontal ligament), accentuated degradation of the alveolar process and partial destruction of cementum. Score 3: accentuated cellular infiltrate, complete resorption of the alveolar process and severe destruction of cementum.

2.8. Measurement of myeloperoxidase activity

The myeloperoxidase (MPO) activity in gingival tissue collected at 6 h after EPD induction was determined as a measurement of neutrophil accumulation. A spectrophotometric assay was utilized to measure MPO activity, as described previously (Bradley et al., 1982). The buccal gingiva surrounding the upper left molars were removed and stored at -70°C . The material was suspended in 0.5% hexadecyltrimethylammonium bromide (HTAB) in 50 mM potassium phosphate buffer, pH 6.0, to solubilize MPO. After homogenized in an ice bath (15 s), the samples were freeze-thawed twice. Additional buffer was added to the test tube to reach 400 μl of buffer per 15 mg of tissue for 12 min. After centrifuging ($1000 \times g/12 \text{ min}$), 0.1 ml of the supernatant was added to 2 ml phosphate buffer 50 mM, pH 6.0, containing 0.167 mg/ml *o*-dianisidine dihydrochloride, distilled water and 0.0005% hydrogen peroxide to give a final volume of 2.1 ml per tube. The absorbance was measured spectrophotometrically (460 nm). One unit of activity was defined as that degrading 1 μmol of peroxide/min at 25°C . Results are expressed in myeloperoxidase units/ml. Staining of smears for MPO activity was performed by the method of Kaplow (1986).

2.9. Microbiological analysis

The buccal gingiva surrounding the upper left molars were removed and placed in 0.3 ml of brain heart infusion broth (BHI). The total transfer time to the microbiology laboratory was less than 1 h. Immediately after, the collected fragment was homogenized and plated in dilutions 1:100 and 1:1000 into blood agar (brain heart infusion agar supplemented with 5% defibrinated sheep blood and henin/menadione 10 $\mu\text{g/ml}$). The microaerophilic environment was obtained using commercially available kits. In order to determine the antimicrobial activity of *Ls* and *Mu*, the samples counts of total microorganisms were done from gingival tissue of rats subjected to periodontitis. Suspected organisms were transferred to brain heart infusion broth and the strains were identified, in order to verify the antimicrobial activity of combined gel, *Streptococcus mutans* (ss-980) and *Candida albicans* ATCC 10239 was tested by established methodology (NCCLS, 2000).

2.10. Detection of IL-1 β and TNF- α in rat maxillary gingival tissue

The buccal gingival tissue from the area surrounding the upper left molars was collected 11 days after periodontitis induc-

tion. The tissue collected was homogenized and processed as described by Safieh-Garabedian et al. (1995). The detection of TNF- α and IL-1 β concentrations was determined by enzyme-linked immunosorbent assay (ELISA) as described previously by Cunha et al. (1993). Briefly, micro titer plates were coated overnight at 4 °C with antibody against rat TNF- α or IL-1 β (10 μ g/ml). After blocking the plates, the samples and standard at various dilutions were added in duplicate and incubated at 4 °C for 24 h. The plates were washed three times with buffer. After washing the plates, 100 μ l of biotinylated sheep polyclonal anti-rat TNF- α or anti-rat IL-1 β (diluted 1/1000 with assay buffer 1% BSA), was added to the wells. After further incubation at room temperature for 1 h, the plates were washed and 100 μ l of avidin-HRP diluted 1:5000 were added. The color reagent *o*-phenylenediamine (OPD; 100 μ l) was added 15 min later and the plates were incubated in the dark at 37 °C for 15–20 min. The enzyme reaction was stopped with H₂SO₄ and absorbance was measured at 490 nm. The results are reported as mean $\pm$ S.E.M. for six animals.

2.11. Statistical analysis

The data are presented as the mean $\pm$ S.E.M. or as the medians. Differences between groups were evaluated by analysis of variance (ANOVA) followed by Bonferroni's test when means were compared or by Kruskal–Wallis test to compare medians. Statistical differences were considered to be significant at $P < 0.05$.

3. Results

3.1. Effect of *Lippia sidoides* (Ls) and *Myracrodruon urundeuva* (Mu) on the alveolar bone loss of the experimental periodontal disease in rats

The treatment of animals subjected to 11 days of experimental periodontal disease with the topical doxycycline gel (10%, w/w, reference substance) or *Myracrodruon urundeuva* 5% and *Lippia sidoides* 0.5% combined based gel significantly ($P < 0.05$) reduced the alveolar bone loss when compared to the vehicle treated group. The groups treated with topical *Myracrodruon urundeuva* 5% (w/w) gel or with topical *Lippia sidoides* 0.5% (v/w) gel, although showing an inhibitory tendency, did not achieve statistically difference from the vehicle treated group (Fig. 1).

3.2. Effect of *Lippia sidoides* (Ls) and *Myracrodruon urundeuva* (Mu) on the histopathological analysis of EPD in rats

The histopathological analysis of the region between the first and second molars of the normal periodontium shows periodontal ligament (Pl), alveolar bone (Ab), cementum (Ce) and gingiva (g) (Fig. 2A). The histopathology of the periodontium of the animals subjected to periodontitis that received vehicle gel (vehicle group) revealed intense inflammatory cell infiltration coupled with cementum destruction and complete alveolar pro-

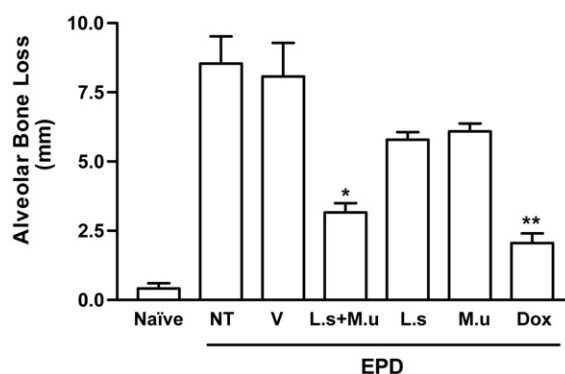


Fig. 1. Effect of *Lippia sidoides* (Ls) and *Myracrodruon urundeuva* (Mu) on the alveolar bone loss in experimental periodontal disease (EPD) in rat: vehicle gel (V), Ls 0.5% (v/w) based gel (L.s), Mu 5% (w/w) based gel (M.u), the combination of Ls 0.5% and Mu 5% gel (L.s+M.u) or doxycycline gel (Dox; 10%, w/w; reference substance) was administered topically, daily, in animals subjected to the EPD induction. EPD was induced also in non-treated animals (NT) and Naïve group received no treatment and was not submitted to EPD induction. Bars represent the mean $\pm$ S.E.M. of alveolar bone loss (mm; $N = 6$). Asterisk indicates a statistically significant difference between treated group and NT group (* $P < 0.05$ and ** $P < 0.01$; Bonferroni, ANOVA).

cess resorption (Fig. 2C), receiving median score 3 (range, 2–3), whereas a reduction of inflammatory cell infiltration and a partial preservation of the cementum and of the alveolar process was found in the periodontium of animals subjected to experimental periodontitis treated with doxycycline gel (10%, w/w; data not shown) and with the topical *Myracrodruon urundeuva* 5% and *Lippia sidoides* 0.5% combined based gel (Fig. 2D), receiving a median score 1 (range, 0–1). These values were statistically significant ($P < 0.05$), when compared to the non-treated and vehicle gel treated groups.

3.3. Effect of *Lippia sidoides* (Ls) and *Myracrodruon urundeuva* (Mu) on the mieloperoxidase activity on gingiva of rats submitted to EPD

The neutrophil infiltration was evaluated by the mieloperoxidase activity in the gingival tissue. A significant ($P < 0.05$) decrease in the mieloperoxidase activity in the gingival tissue was observed only in the group treated with the topical *Myracrodruon urundeuva* 5% and *Lippia sidoides* 0.5% combined based gel, as compared to vehicle-treated group (Fig. 3).

3.4. Effect of *Lippia sidoides* (Ls) and *Myracrodruon urundeuva* (Mu) on the gingiva IL-1 β and TNF- α concentrations of rats submitted to EPD

The treatment with topical *Myracrodruon urundeuva* 5% and *Lippia sidoides* 0.5% combined based gel significantly ($P < 0.05$) decreased the cytokine (TNF- α , Fig. 4A and IL-1 β , Fig. 4B) levels in gingival tissue of rats submitted to EPD, when compared to non-treated group or to vehicle gel treated group. The groups treated with topical *Myracrodruon urundeuva* 5% (w/w) gel or with topical *Lippia sidoides* 0.5% (v/w) gel did not achieve statistically difference from the vehicle treated group (Fig. 4A and B).

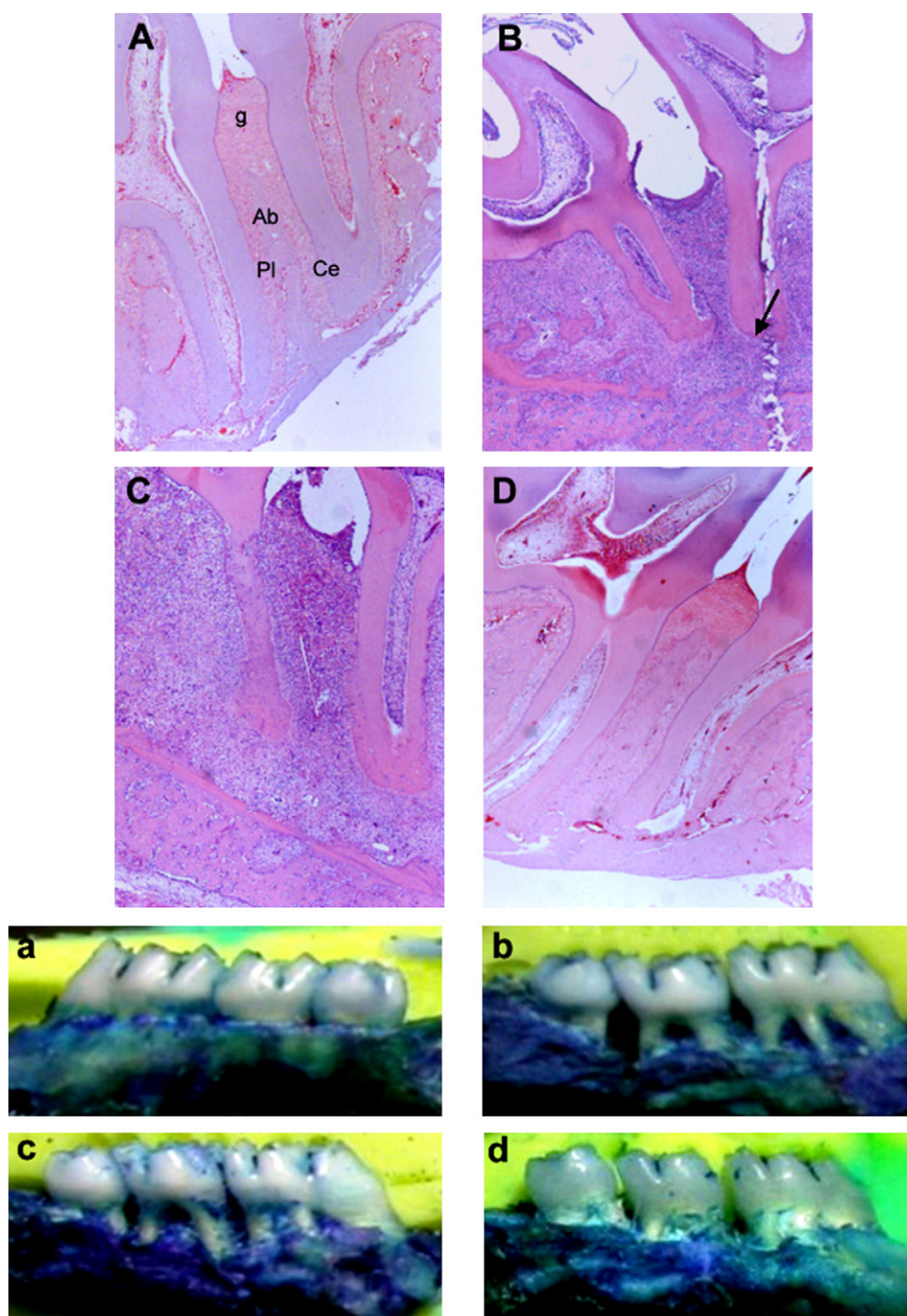


Fig. 2. Histopathological and macroscopic aspects of the effect of *Lippia sidoides* (*Ls*) and *Myracrodruon urundeuva* (*Mu*) on the experimental periodontal disease (EPD) in rat. (Panel A) Microscopic aspect of a normal periodontium of rat maxillae showing gingiva (g), periodontal ligament (Pl), cementum (Ce), alveolar bone (Ab). (Panel B) Periodontium of rat subjected to experimental periodontitis with no treatment showing severe inflammatory cell infiltration in both gingiva and periodontal ligament, cementum destruction (arrow) and total resorption of the alveolar process. (Panel C) Periodontium of rat subjected to experimental periodontitis and treated with vehicle gel showing alterations similar to non-treated rats. (Panel D) Periodontium of rat subjected to experimental periodontitis and treated with *Ls* + *Mu* gel showing absence of inflammatory cell infiltration and preservation of cementum and alveolar bone (hematoxylin and eosin stain; magnification $\times 40$). (Panel a) Macroscopic aspects of a normal maxillae. (Panel b) Macroscopic aspects of the maxillae excised from a non-treated animal subjected to experimental periodontitis showing severe bone resorption and root exposure. (Panel c) Macroscopic aspects of the maxillae excised from an animal subjected to experimental periodontitis and treated with vehicle gel with little reduction in bone resorption. (Panel d) Macroscopic aspects of the maxillae excised from an animal treated with *Ls* + *Mu* showing significant reduction in bone resorption.

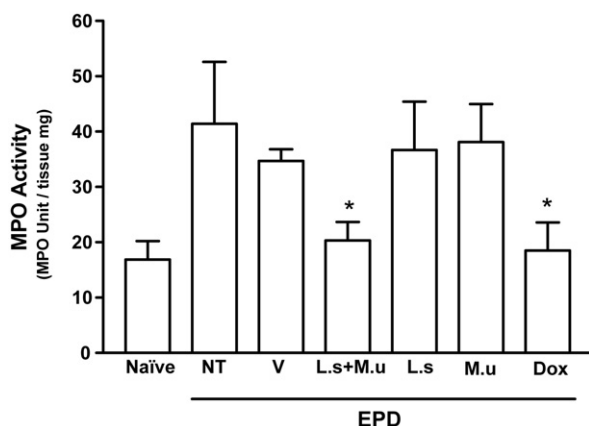


Fig. 3. Effect of *Lippia sidoides* (Ls) and *Myracrodruon urundeuva* (Mu) on myeloperoxidase (MPO) activity in the maxillary gingival tissue of rats submitted to experimental periodontal disease (EPD). Vehicle gel (V), Ls 0.5% (v/w) based gel (L.s), Mu 5% (w/w) based gel (M.u), the combination of Ls 0.5% and Mu 5% gel (L.s + M.u) or doxycycline gel (Dox; 10%, w/w; reference substance) was administered topically in animals submitted to EPD induction. EPD was induced also in non-treated animals (NT) and Naïve group received no treatment and was not submitted to EPD induction. Bars represent mean $\pm$ S.E.M. of the activity of MPO/mg of tissue. * $P < 0.05$ was considered significantly different compared to NT group (ANOVA; Bonferroni's test).

3.5. Antibacterial effect of *Lippia sidoides* (Ls) and *Myracrodruon urundeuva* (Mu) in oral microorganisms of rats submitted to EPD

The microbiological analysis in the gingival tissue of rats subjected to the EPD showed that the groups either treated with topical *Myracrodruon urundeuva* 5% (w/w) gel or with topical *Lippia sidoides* 0.5% (v/w) gel as well as the group treated with topical *Myracrodruon urundeuva* 5% and *Lippia sidoides* 0.5% combined based gel showed a significant ($P < 0.05$) decrease of periodontal microorganisms growth (Fig. 5). The group treated with topical doxycycline based gel (10%, w/w; reference substance) also showed an inhibition upon periodontal microorganisms growth.

3.6. Effect of *Lippia sidoides* (Ls) and *Myracrodruon urundeuva* (Mu) on the weight loss of rats submitted to EPD

The weight loss, which is characteristically seen in the animals subjected to the experimental periodontitis (Lima et al., 2000), was significantly ($P < 0.05$) prevented by the treatment with topical *Myracrodruon urundeuva* 5% (w/w) gel or with topical *Lippia sidoides* 0.5% (v/w) gel as well as the group treated with topical *Myracrodruon urundeuva* 5% and *Lippia sidoides* 0.5% combined based gel. The group treated with topical doxycycline based gel (10%, w/w; reference substance) also had weight loss prevented (Fig. 6). No signs of skin irritation or redness were observed during the 11 days treatment.

4. Discussion

In the present study, we have shown that the local application of gel based on fluid extract from *Myracrodruon urundeuva*

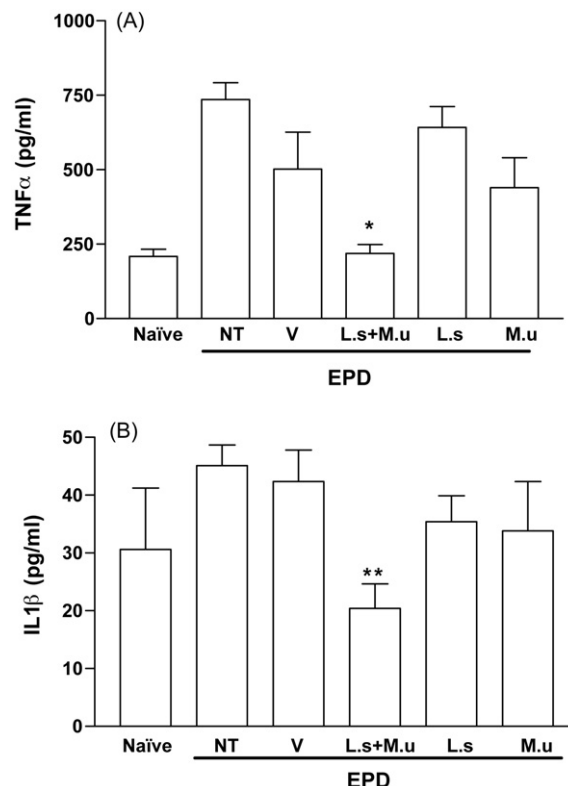


Fig. 4. Effect of *Lippia sidoides* (Ls) and *Myracrodruon urundeuva* (Mu) on TNF- α and interleukin-1 β production in the buccal gingiva of rats submitted to experimental periodontitis disease (EPD). Vehicle gel (V), Ls 0.5% (v/w) based gel (L.s), Mu 5% (w/w) based gel (M.u) or the combination of Ls 0.5% and Mu 5% gel (L.s + M.u) was administered topically in animals submitted to the EPD. EPD was induced also in non-treated animals (NT) and Naïve group received no treatment and was not submitted to EPD induction. After 11 days, the animals were sacrificed and the buccal gingiva from left upper molar region was removed and the concentrations of TNF- α (panel A), interleukin-1 β (IL-1 β , panel B) in the homogenate supernatant were determined by ELISA. Results are reported as mean $\pm$ S.E.M. of six wells (1 well per animal) and asterisk indicates a statistically significant difference between treated groups and NT group (* $P < 0.05$; Bonferroni, ANOVA).

combined with essential oil from *Lippia sidoides* plays a positive role on the bone loss associated with experimental periodontal disease. This effect was associated with reduction of the inflammatory reaction and antimicrobial activity.

The gel containing extracts from both plants have shown more intensive anti-inflammatory action than their isolated applications, inhibiting neutrophil infiltration as early as 6 h after its EPD induction as confirmed by the reduction of MPO activity on the gingival tissue. This effect was associated with a reduction in the chronic inflammatory cell influx on rat gingival tissue and periodontal ligament seen clearly on the histopathology on the 11th day after induction of EPD.

In this study we selected a two-fold concentration of both compounds (*Lippia sidoides* 0.5% and *Myracrodruon urundeuva* 5%) based on our pilot study. These data are consistent with the previous reports that described antiplaque effect of *Lippia sidoides* EO (Botelho and Soares, 1994; Botelho, 2005); the colonic anastomotic healing effect in rats that received 10% Aroeira-do-sertao (*Myracrodruon urundeuva* Fr. All.) by

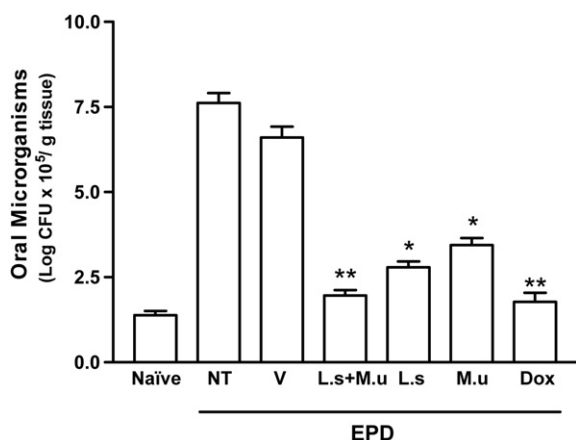


Fig. 5. Antibacterial effect of *Lippia sidoides* (Ls) and *Myracrodruon urundeuva* (Mu) in bacteria involved in experimental periodontal disease (EPD) in rats. Vehicle gel (V), Ls 0.5% (v/w) based gel (L.s), Mu 5% (w/w) based gel (M.u), the combination of Ls 0.5% and Mu 5% gel (L.s + M.u) or doxycycline gel (Dox; 10%, w/w; reference substance) was administered topically in animals submitted to the EPD. EPD was induced also in non-treated animals (NT). Results are reported as mean + S.E.M. of the colonies forming units (CFU) count/g of tissue from the rats subjected to EPD. Asterisk indicates a statistically significant difference between treated groups and NT group (* $P < 0.05$ and ** $P < 0.01$; Bonferroni, ANOVA).

enema; the inhibitory effects of tannin fraction extracted from the bark of *Myracrodruon urundeuva* on neutrophil migration to peritoneal cavity as well as vesical edema and increased vascular permeability in hemorrhagic cystitis induced by cyclophosphamide (Viana et al., 1997); and the suppressive effect of chalcone-enriched fraction on neutrophil dependent paw edema induced by carrageenan (Viana et al., 2003). Neutrophils are important cells in host defense against infecting bacteria. However, these cells have also been linked to tissue destruction. The

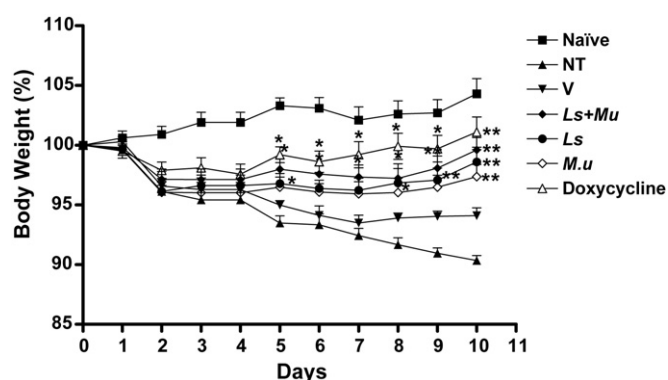


Fig. 6. Effect of *Lippia sidoides* (Ls) and *Myracrodruon urundeuva* (Mu) on weight changes in rats submitted to the experimental periodontitis disease (EPD). Vehicle gel (V), Ls 0.5% (v/w) based gel (L.s), Mu 5% based gel (M.u), the combination of Ls 0.5% and Mu 5% gel (L.s + M.u) or doxycycline gel (Dox; 10%, w/w; reference substance) was administered topically in animals submitted to the EPD. EPD was induced also in non-treated animals (NT) and Naïve group received no treatment and was not submitted to EPD induction. Data represent mean $\pm$ S.E.M. ($N = 6$) of the weight (g) measured each day before and after the EPD induction until the 11th day. Asterisk indicates a statistically significant difference between treated groups and NT group (* $P < 0.05$ and ** $P < 0.01$; ANOVA, Bonferroni's test).

importance of the acute neutrophil infiltration on gingival tissue in the evolution of periodontal disease has been demonstrated before (Liu and Pope, 2000; Menezes et al., 2005). The reduction of neutrophil influx into gingival tissue as confirmed by MPO activity could be an explanation for the protective effect of the combined gel on the periodontal tissue including bone and cementum.

In the present study it was also shown a reduction of levels of pro-inflammatory cytokines such as TNF- α and IL-1 β on the gingival tissue of animals treated with gel containing association of *Myracrodruon urundeuva* and *Lippia sidoides*. In agreement with our findings, chalcones have been accounted for reduction of inducible nitric oxide synthase (iNOS) protein expression, nitric oxide and TNF- α production. The importance of inflammatory cytokines, particularly TNF- α and IL-1 β , on periodontal disease has been demonstrated previously (Rossomando and White, 1993; Lima et al., 2000, 2004). TNF may potentiate periodontitis by stimulating the release of eicosanoids and other cytokines, such as IL-1 and TNF- α . IL-1 activates neutrophils and macrophages, increasing the production and the release of reaction oxygen species and nitric oxide, which has been implicating in local tissue damage (Assuma et al., 1998). Thus, the inhibition of these cytokines by the combined gel could be contributing to the reduction of neutrophil infiltration, bone and cementum destruction.

Either the gel containing fluid extract from *Myracrodruon urundeuva* combined with essential oil from *Lippia sidoides* or the gels containing the *Myracrodruon urundeuva* or *Lippia sidoides* presented significant antibacterial activity as shown by the counting of the periodontal bacteria. Bacteria are considered to be the primary etiologic agents of periodontal disease (Ximenez-Fyvie et al., 2000). Study from our group demonstrated that the periodontal microbial flora change during periodontal disease in rats, so that anaerobic Gram-negative bacilli predominate (Menezes et al., 2005). The antibacterial and antifungal properties of *Lippia sidoides* essential oil have been demonstrated (Botelho et al., 2007; Fontenelle et al., 2007), which are largely attributed to the presence of thymol and carvacrol. In accordance with our data, a double-blind clinical trial has demonstrated that a mouth-rinse of *Lippia sidoides* EO decreases salivary *Streptococcus mutans* levels as well the clinical scores of gingival health in humans as compared to 0.12% chlorhexidine (Botelho, 2005). In addition, study from our group demonstrated that essential oil from *Lippia sidoides*, thymol and carvacrol has shown an antibacterial activity against oral pathogens including *Streptococcus sanguis*, *Streptococcus mutans* and *Candida albicans* (Botelho et al., 2007). The gel containing *Myracrodruon urundeuva* also presented antibacterial effect. A non-specific effect of local application can be discharged by the fact that the control gel did not reduce the bacterial counting. Thus, the inhibition of bacterial growth as a result of the treatment with *Myracrodruon urundeuva* and/or *Lippia sidoides* could be related to the reduction on MPO activity, cytokine release and bone and cementum destruction and could be accounted for the synergistic effect between *Myracrodruon urundeuva* and *Lippia sidoides* on these parameters.

5. Conclusions

This study demonstrates that the local application of dental gel based on *Lippia sidoides* (0.5%) (*alecrim pimenta*) and/or *Myracrodruon urundeuva* (5%) (*aroeira-do-sertao*) has anti-inflammatory and antimicrobial effects and prevents alveolar bone resorption and the weight loss in experimental periodontitis. The study revealed the great potential of the combined gel preparation to combat periodontal disease and further studies are warranted to clarify its usefulness in clinical situations.

Acknowledgements

This work was supported by FUNCAP Fellowship Grant (#9131/06) and CNPq the Brazilian Agencies for Scientific and Technological Development.

The authors thank Luciano Lima Correa Department of Community Health and Luiz Odorico Monteiro (Municipal Secretary of Health).

Maria Silvandira França Pinheiro, Department of Physiology and Pharmacology, and José Ivan Rodrigues de Sousa, Department of Morphology, Faculty of Medicine, Federal University of Ceará, Brazil, for technical assistance.

References

- Assuma, R., Oates, T., Cochran, D., Amar, S., Graves, D.T., 1998. IL-1 and TNF antagonists inhibit the inflammatory response and bone loss in experimental periodontitis. *Journal of Immunology* 160, 403–409.
- Botelho, M.A., 2005. Eficácia do alecrim pimenta (*Lippia sidoides*) no controle da placa bacteriana e gengivite: Um ensaio clínico duplo cego controlado e randomizado. Tese apresentada ao Curso de Mestrado em Saúde Pública da Universidade Federal do Ceará, Fortaleza-Ce, Brasil, p. 113.
- Botelho, M.A., Nogueira, N.A.P., Bastos, G.M., Fonseca, S.G.C., Lemos, T.L.G., Matos, F.J.A., Montenegro, D., Heukelbach, J., Rao, V.S., Brito, G.A.C., 2007. Antimicrobial activity of the essential oil from *Lippia sidoides*, carvacrol and thymol against oral pathogens. *Brazilian Journal of Medical and Biological Research* 40, 349–356.
- Botelho, M.A., Soares, J.B., 1994. O óleo essencial do alecrim pimenta (*Lippia sidoides* Cham.) na prevenção da cárie dental. In: XIII Simpósio de Plantas Mediciniais do Brasil, Fortaleza, p. 20.
- Bradley, P.P., Christensen, R.D., Rothstein, G., 1982. Cellular and extracellular myeloperoxidase in pyogenic inflammation. *Blood* 60, 618–622.
- Chambrone, L.A., Chambrone, L., 2006. Tooth loss in well-maintained patients with chronic periodontitis during long-term supportive therapy in Brazil. *Journal of Clinical Periodontology* 33, 759–764.
- Craveiro, A.A., Matos, F.J.A., Alencar, J.W., 1976. A simple and inexpensive steam generator for essential oils extraction. *Journal of Chemical Education* 53, 652.
- Crawford, J.M., Taubman, M.A., Smith, D.J., 1978. The natural history of periodontal bone loss in germfree and gnotobiotic rats infected with periodontopathic microorganisms. *Journal of Periodontal Research* 13, 316–325.
- Cunha, F.Q., Boukili, M.A., Motta, J.I.B., Vargaftig, B.B., Ferreira, S.H., 1993. Blockade by fenspiride of endotoxin-induced neutrophil migration in the rat. *European Journal of Pharmacology* 238, 47–52.
- Fontenelle, R.O., Morais, S.M., Brito, E.H., Kerntopf, M.R., Brilhante, R.S., Cordeiro, R.A., Tome, A.R., Queiroz, M.G., Nascimento, N.R., Sidrim, J.J., Rocha, M.F., 2007. Chemical composition, toxicological aspects and antifungal activity of essential oil from *Lippia sidoides* Cham. *Journal of Antimicrobial Chemotherapy* 59, 934–940.
- Goes, A.C., Rodrigues, L.V., de Menezes, D.B., Grangeiro, Mdo.P., Cavalcante, A.R., 2005. Histologic analysis of colonic anastomotic healing, in rats, under the action of 10% Aroeira-do-sertao (*Myracrodruon urundeuva* Fr. All.) enema. *Acta Cirurgica Brasileira* 20, 144–151.
- Kaplow, L.S., 1986. The relationship of myeloperoxidase activity to neutrophil maturity and other hematologic indicators of infection. *Blood Cells* 12, 153–165.
- Lacoste, E., Chaumont, J.P., Mandin, D., Plumel, M.M., Matos, F.J.A., 1996. Antiseptic properties of essential oil of *Lippia sidoides* Cham. Application to the cutaneous microflora. *Annales Pharmaceutiques Françaises* 54, 228–230.
- Leitão, R.F., Ribeiro, R.A., Chaves, H.V., Rocha, F.A., Lima, V., Brito, G.A., 2005. Nitric oxide synthase inhibition prevents alveolar bone resorption in experimental periodontitis in rats. *Journal of Periodontology* 76, 956–963.
- Lemos, T.L.G., Craveiro, A.A., Alencar, J.W., Matos, F.J.A., Clark, A.M., Machesney, J.D., 1990. Antimicrobial activity of essential oil of Brazilian plants. *Phytotherapy Research* 4, 82–84.
- Lima, V., Bezerra, M.M., Alencar, V.B.M., 2000. Effects of chlorpromazine on alveolar bone loss in experimental periodontal disease in rats. *European Journal of Oral Science* 108, 123–129.
- Lima, V., Vidal, F.D., Rocha, F.A., Brito, G.A., Ribeiro, R.A., 2004. Effects of tumor necrosis factor-alpha inhibitors pentoxifylline and thalidomide on alveolar bone loss in short-term experimental disease in rats. *Journal of Periodontology* 75, 162–168.
- Liu, H., Pope, R.M., 2000. Phagocytes: mechanisms of inflammation and tissue destruction. *Rheumatic Disease Clinics of North America* 30, 19–39.
- Matos, F.J.A., 2000. Plantas Mediciniais. Imprensa Universitária, 2ª Edição, Fortaleza, Brazil, p. 344.
- Menezes, A.M., Rocha, F.A., Chaves, H.V., Carvalho, C.B., Ribeiro, R.A., Brito, G.A., 2005. Effect of sodium alendronate on alveolar bone resorption in experimental periodontitis in rats. *Journal of Periodontology* 76, 1901–1909.
- Monteiro, J.M., de Albuquerque, U.P., Lins-Neto, E.M., de Araújo, E.L., de Amorim, E.L., 2006. Use patterns and knowledge of medicinal species among two rural communities in Brazil's semi-arid northeastern region. *Journal of Ethnopharmacology* 105, 173–186.
- Monteiro, M.V., de Melo Leite, A.K., Bertini, L.M., de Morais, S.M., Nunes Pinheiro, D.C., 2006. Topical anti-inflammatory, gastroprotective and antioxidant effects of the essential oil of *Lippia sidoides* Cham. leaves. *Journal of Ethnopharmacology* 111, 378–382.
- NCCLS (National Committee for Clinical Laboratory Standards), 2000. Methods for Dilution Antimicrobial Susceptibility Test for Bacteria that Grow Aerobically, fifth ed. Approved Standard, Wayne, PA, p. M7-A5.
- Rossomando, E.F., White, L., 1993. A novel method for the detection of TNF-alpha in gingival crevicular fluid. *Journal of Periodontology* 64, 445–449.
- Safieh-Garabedian, B., Poole, S., Allchorne, A., Winter, J., Woolf, C.J., 1995. Contribution of interleukin-1 beta to the inflammation-induced increase in nerve growth factor levels and inflammatory hyperalgesia. *British Journal of Pharmacology* 115, 1265–1275.
- Sallay, K., Sana, vi.F., Ring, I., Pham, P., Behling, U.H., Nowotny, A., 1982. Alveolar bone destruction in the immuno-suppressed rat. *Journal of Periodontal Research* 17, 263–274.
- Samejima, V., Ebisu, S., Okada, H., 1990. Effect of injection with *Eikenella corrodens* on the progression of ligature induced periodontitis in rats. *Journal of Periodontal Research* 25, 308–315.
- Viana, G.S., Bandeira, M.A., Matos, F.J.A., 2003. Analgesic and antiinflammatory effects of chalcones isolated from *Myracrodruon urundeuva* Allemão. *Phytomedicine* 10, 189–195.
- Viana, G.S., Bandeira, M.A., Moura, L.C., Souza-Filho, M.V.P., Matos, F.J.A., Ribeiro, R.A., 1997. Analgesic and antiinflammatory effects of the tannin fraction from *Myracrodruon urundeuva* Fr. All. *Phytotherapy Research* 11, 118–122.
- Ximenez-Fyvie, L.A., Haffajee, A.D., Socransky, S.S., 2000. Comparison of the microbiota of supra and subgingival plaque in health and periodontitis. *Journal of Clinical Periodontology* 27, 648–657.